Validity of the Kinarm Approach-Avoidance Task for Assessing Automatic Tendencies Towards Physical Activity and Sedentary Behaviours: A Registered Report

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ABSTRACT

Background. Approach-avoidance tendencies have traditionally been assessed using unidimensional reaction times, measured by moving an on-screen manikin towards or away from stimuli using a keyboard, joystick, touchscreen, or mouse. Recently, a bimanual robotic task has been proposed to provide a continuous, high-resolution, multidirectional, and ecologically relevant assessment of the spatiotemporal dynamics that characterize these tendencies.

Objective. To assess the construct, convergent, concurrent, and predictive validity of the approach-avoidance task performed on the Kinarm robotic platform with stimuli depicting physical activity and sedentary behaviors.

Methods. Healthy participants performed the Kinarm approach-avoidance task and the keyboard-based manikin task. Approach-avoidance tendencies were quantified using reaction time, time to peak velocity, x-flips, and opposite displacement. Physical activity was assessed with accelerometers. Additional questionnaires measured intentions, explicit attitudes, automaticity, and the valuation of physical effort. Trial-level approach-avoidance tendencies were analyzed using linear mixed-effects models. Convergent validity between the Kinarm and manikin tasks and predictive associations with physical activity and related psychological constructs were evaluated using correlations and hierarchical regressions, controlling for age, sex, height, weight, handedness, error rate, and elapsed time.

Keywords. Psychology, Motivation, Robotics, Validation Study, Exercise, Sedentary Behaviour

INTRODUCTION

Automatic motivational processes have been shown to influence health behaviours (1,2), including physical activity behaviour (3–7). These processes operate outside conscious awareness, are shaped by past experiences, and are spontaneously triggered by contextual cues (8). Among these processes, approach-avoidance tendencies are thought to be particularly proximal to behaviour, as they reflect automatic action orientations towards or away from environmental stimuli (1,9–11).

In health psychology, approach-avoidance tendencies have traditionally been assessed using unidimensional reaction times to task-relevant stimuli, measured via interfaces such as a keyboard, joystick, touchscreen, or computer mouse (12–16). Consistent with this broader literature, physical activity research has primarily assessed approach-avoidance tendencies using the manikin task in which reaction times are measured by moving an on-screen manikin towards (i.e., approach) or away (i.e., avoid) from stimuli depicting physical activity and sedentary behaviors (3–5,11,17–31).

The use of reaction-time paradigms, such as the manikin task, is theoretically grounded in dual-process models of physical activity (32,33). These models posit that, through repeated experiences, the concept of physical activity can become associated with positive or negative affective responses (i.e., pleasure or displeasure). As a consequence of these learned associations, environmental cues related to physical activity automatically elicit an affective valuation, which biases the activation of approach or avoidance impulses and thereby facilitates or inhibits the planning and initiation of actions towards these cues (e.g., approaching a positively valued stimulus and avoiding a negatively valued stimulus). By measuring the time required to initiate conceptual (i.e., via keyboard) or actual (e.g., via joystick) approach or avoidance movements in response to stimuli related to physical activity, reaction times provide an indirect measure of the strength and direction of automatically activated tendencies.

However, despite their theoretical grounding and widespread use, unidimensional reaction-time measures have limitations. Because reaction times collapse the motor response into a single timepoint, they conflate multiple perceptual, cognitive, and motor processes and overlook the temporal unfolding and spatial characteristics of the response. In addition, constraining responses to a single spatial dimension (e.g., anteroposterior joystick movements) may oversimplify the inherently multidirectional nature of approach-avoidance tendencies and limit sensitivity to how these tendencies are expressed across space.

Other limitations relate to the task implementation. In joystick-based tasks, approach (i.e., pulling) and avoidance (i.e., pushing) are often simulated via stimulus zooming in and out, respectively, which introduces ambiguity because the same response can be interpreted differently. For example, pulling a joystick brings the arm away from the screen and stimulus (i.e., withdrawal), yet it is typically paired with visual stimulus enlargement, which conveys approach. This mismatch may confound the interpretation of reaction-time differences (22). By relying on symbolic and discretized responses, keyboard-based tasks such as the manikin task avoid this reference-frame ambiguity. However, these key presses only indirectly reflect bodily action, these tasks introduce limitations in ecological validity and sensitivity to fine-grained temporal and spatial variations in responses.

Taken together, these limitations highlight the need for paradigms that capture approach-avoidance tendencies as a dynamic and embodied process rather than as a single latency measure constrained to one spatial dimension. To address this need, a bimanual robotic task has been proposed that provides a continuous, high-resolution, multidirectional, and ecologically relevant

assessment of the spatiotemporal dynamics that characterize approach-avoidance tendencies (34). However, this task, implemented on the Kinarm robotic platform and referred to as the Kinarm Approach-Avoidance Task (KAAT), has not yet undergone formal psychometric validation. Accordingly, the objective of this study is to evaluate the construct, convergent, concurrent, and predictive validity of the KAAT.

Hypotheses

We formulated hypotheses (H) corresponding to each of the four types of validity addressed by our objective.

Construct Validity

To assess the structural aspect of construct validity, which examines whether a measure follows expected internal patterns, we hypothesise that raw reaction times in the KAAT will differ between the approach and avoidance conditions for both physical activity (H1a) and sedentary stimuli (H1b). We further hypothesise that relative bias derived from the KAAT, which accounts for interindividual differences in generic approach-avoidance tendencies by accounting for reaction times to neutral stimuli (11), will significantly differ from zero for both physical activity (H1c) and sedentary stimuli (H1d).

To assess the sensitivity aspect of construct validity, which examines whether a measure demonstrates enhanced sensitivity to the underlying construct as theoretically expected, we hypothesise that approach-avoidance differences in reaction times, reflecting raw bias, will be larger when assessed with the KAAT than with the manikin task for both physical activity (H1e) and sedentary stimuli (H1f). We formulated analogous hypotheses for relative bias (H1g, H1h).

Convergent Validity

To assess convergent validity, which examines the extent to which different measures intended to assess the same construct are related, we hypothesize that raw bias will show small positive correlations between tasks for both physical activity (H2a) and sedentary stimuli (H2b), consistent with previous studies showing limited convergence between approach-avoidance tasks (13,35). We formulated analogous hypotheses for relative bias (H2c, H2d).

Concurrent Validity

To assess concurrent validity, which examines the extent to which a measure is associated with theoretically related constructs assessed at the same time, we hypothesise that the relative bias towards physical activity derived from the KAAT will be associated with (H3a) device-based total physical activity, (H3b) device-based moderate-to-vigorous physical activity, (H3c) explicit affective attitudes towards physical activity, (H3d) intentions to be physically active, (H3e) valuation of physical effort, and (H3f) automaticity towards physical activity. To compare the concurrent validity of the KAAT and the manikin task, we hypothesise that these associations will be stronger for the KAAT than for the manikin task (H3g-l).

Predictive Validity

To assess predictive validity, which refers to the extent to which a measure predicts relevant behavioural outcomes, we hypothesise that relative bias towards physical activity stimuli measured with the KAAT will explain variance in device-based total physical activity (H4a), and that kinematic variables (i.e., time to peak velocity, x-flips, and opposite displacement) will

explain additional variance beyond that accounted for by relative bias of the KAAT (H4b). We further hypothesise that the relative bias towards physical activity stimuli measured with the KAAT will explain more variance than when assessed with the manikin task (H4c).

Table 1. A priori hypotheses by validity type

Construct Validity (H1)		
Internal Structure		
H1a: KAAT raw $RT_{avoid\ PA}$ ≠ KAAT raw $RT_{approach\ PA}$	**H1c**: KAAT relative $Bias_{PA}$ ≠ 0	
H1b: KAAT raw $RT_{avoid\ SED}$ ≠ KAAT raw $RT_{approach\ SED}$	**H1d**: KAAT relative $Bias_{SED}$ ≠ 0	
Sensitivity		
H1e: |KAAT raw $Bias_{PA}$|>|Manikin raw $Bias_{PA}$|	**H1g**: |KAAT relative $Bias_{PA}$| > |Manikin relative $Bias_{PA}$|	
H1f: |KAAT raw $Bias_{SED}$|>|Manikin raw $Bias_{SED}$|	**H1h**: |KAAT relative $Bias_{SED}$| > |Manikin relative $Bias_{SED}$|	
Convergent Validity (H2)		
H2a: KAAT & Manikin raw $Bias_{PA}$ are associated	**H2c**: KAAT & Manikin relative $Bias_{PA}$ are associated	
H2b: KAAT & Manikin raw $Bias_{SED}$ are associated	**H2d**: KAAT & Manikin relative $Bias_{SED}$ are associated	
Concurrent Validity (H3)		
KAAT relative $Bias_{PA}$ is associated with:		
H3a: Device-based total PA	**H3c**: Explicit attitudes towards PA	**H3e**: Valuation of physical effort
H3b: Device-based moderate-to-vigorous PA	**H3d**: Intentions to be PA	**H3f**: Automaticity towards PA
KAAT relative $Bias_{PA}$ is more strongly associated than Manikin relative $Bias_{PA}$ with:		
H3g: Device-based total PA	**H3i**: Explicit attitudes towards PA	**H3k**: Valuation of physical effort
H3h: Device-based moderate-to-vigorous PA	**H3j**: Intentions to be PA	**H3l**: Automaticity towards PA
Predictive Validity (H4)		
H4a: KAAT relative $Bias_{PA}$ explains variance in device-based total PA		
H4b: Kinematic variables explain additional variance in total PA beyond that accounted for by KAAT relative $Bias_{PA}$		
H4c: Relative $Bias_{PA}$ better explains total PA when assessed with the KAAT than with the manikin task		

Notes. KAAT = Kinarm approach avoidance task, RT = reaction time, H = hypothesis, PA = physical activity, SED = sedentary behavior

METHODS

Participants

Participants were recruited via social media, emails, posters, and word of mouth. Inclusion criteria were age between 18 and 60 years and the ability to communicate in French or English. Exclusion criteria included any conditions that could influence our results, such as self-reported depression, upper or lower limb injury, a history of psychiatric or neurological disorders, alcohol or drug abuse, or the use of psychotropic medications or illicit substances that may affect attention or neurological functioning. The study was approved by the University of Ottawa Research Ethics Board (H-01-25-11075). All participants provided informed consent and were not compensated for their participation. Data were collected between April 2025 and February 2026. Although 100% of the data have been collected, no observations, preprocessing, or analyses have taken or will take place prior to stage 1 in-principle acceptance, which corresponds to Level 3 of bias control according to the Peer Community In Registered Report taxonomy.

Power Analysis

Simulation-Based Power Analysis for Linear Mixed-Effects Models

Because the interaction effects, such as relative bias, are generally more difficult to detect than simple effects, such as raw bias, we powered the study to detect this interaction. This approach provides a conservative sample-size estimate for H1a and H1b while directly targeting the primary effect interest for H1c and H1d. In linear mixed-effects models, classical power analyses often yield biased or overly optimistic estimates because they do not account for hierarchical variance

components and item-level variability (36,37). Therefore, our primary power analysis was conducted using Monte-Carlo simulations with the *simr* package in R (38), which is specifically designed for power estimation in mixed-effects models.

Because no directly comparable effect size estimates were available in the literature for the effects under investigation here, we derived the model parameters from an open dataset (39) obtained from a prior study (4) that used the same manikin task structure and stimuli as in our study. Identical exclusion criteria to those planned for the present study were applied. The dataset included 29 participants, 192 trials per participant, and 30 pictograms (5 depicting physical activity, 5 depicting sedentary behavior, 20 neutral). For each experimental stimulus category (physical activity or sedentary behavior), we fitted the following linear mixed-effects model using maximum likelihood estimation with deviation coding (−0.5/+0.5): Reaction Time ~ Movement × Stimulus Type + (1 + Movement | Subject) + (1 | Pictogram). Movement reflected the required response direction (approach vs. avoidance), whereas Stimulus Type contrasted experimental stimuli (physical activity or sedentary pictograms) with neutral pictograms. The selected random-effects structure corresponded to the most theoretically justified specification, including by-subject random slopes for Movement (40), while remaining identifiable and free of singular fits. Consistent with current recommendations to balance model complexity, statistical power, and numerical stability (41,42), we retained the most complex convergent structure supported by the data. Fixed-effect estimates, random-effect variance components (VarCorr), and residual variance (σ^2) from these pilot models were then used to generate simulated datasets reproducing the planned balanced experimental design for each experimental stimulus category (96 neutral trials and 48 experimental trials per participant, with equal numbers of approach and avoidance trials; 20 neutral and 5 experimental stimuli sampled within participants). Statistical power for the Movement × Stimulus Type interaction was estimated using 1000 Monte-Carlo simulations ($\alpha = .05$; t-test with Satterthwaite-approximated degree of freedom on the fixed effect), yielding a Monte-Carlo standard error below ~1% for the power values around .90 (43). We computed simulation-based power curves using the *powerCurve* function. These curves indicated that achieving 90% power required 32 participants for physical activity stimuli and 45 participants for sedentary stimuli (Figure 1). We therefore set 45 participants as the minimum required sample size. However, because the simulation-based power analysis relies on variance components and fixed-effect estimates derived from a single prior dataset and thus carries parameter uncertainty (44), recruitment will continue until the pre-specified end of data collection in February 2026. This time-based stopping reduces the risk of an underpowered study if the true effects are smaller than anticipated and ensures sensitivity for the correlational and regression hypotheses (H2–H4).

Sensitivity Power Analysis

Powering the two-way Movement × Task interaction testing H1e and H1f and the three-way Movement × Stimulus Type × Task interaction testing H1g and H1h, where Task contrasts the KAAT and the manikin task, was not feasible because the KAAT is novel and reliable estimates for between-task differences are unavailable. Therefore, these interactions will be interpreted using effect sizes and confidence intervals rather than relying solely on null-hypothesis significance testing.

For H2a-d, sensitivity power analysis was conducted using G*Power for a bivariate correlation exact test, assuming a two-tailed $\alpha = .05$, 80% power, and a total sample size of N = 90, corresponding to the number of participants that can feasibly be reached. This analysis yielded a smallest detectable correlation of $|r| = .29$.

For H3a-f, sensitivity power analyses were designed to align directly with the planned hypothesis tests, which evaluate whether relative bias derived from the KAAT was associated with each outcome. The regression models included eight predictors: age, sex, height, weight, handedness, total number of errors, time on the task, and the KAAT relative bias term, with the latter being the sole predictor of interest. Sensitivity was estimated as the additional variance explained (ΔR^2) by the KAAT relative bias term when added to the model.

For H3g-l, the analyses evaluate whether the relative bias derived from the KAAT is more strongly associated with each outcome than the relative bias derived from the manikin task. The two task-specific relative biases were reparametrized into their mean (capturing shared variance across tasks) and their difference (KAAT – Manikin). This difference term is algebraically equivalent to the planned coefficient contrast ($\beta_{KAAT} - \beta_{Manikin}$) and therefore directly represents the task-difference effect of interest. Each regression model included nine predictors: age, sex, height, weight, handedness, total number of errors, time on the task, the mean bias term, and the bias difference term, with the latter specified as the sole predictor of interest. Sensitivity was estimated as the additional variance explained (ΔR^2) by the bias difference term when added to the model.

Under these assumptions ($\alpha = .05$, power = .80, N = 90), a priori sensitivity analyses were conducted using G*Power (F tests; linear multiple regression: fixed model, R^2 increase) for H3a-l. The incremental effect was expressed as Cohen's f^2, which can be used to compute $\Delta R^2 = f^2(1 - R^2_{full})$. The smallest detectable incremental effect was $f^2 = 0.089$, which corresponds to a detectable increase in explained variance of approximately $\Delta R^2 = 0.05–0.08$ when assuming plausible full-model R^2 values between 0.10 and 0.40. Because ΔR^2 equals the squared semi-partial correlation of the predictor of interest, this corresponds to a detectable semi-partial association of approximately r = 0.23–0.28. In the absence of strong multicollinearity among predictors, standardized regression coefficients (β) are expected to be of comparable magnitude. Accordingly, standardized coefficients of $|\beta_{KAAT}|$ or $|\beta_{KAAT} - \beta_{Manikin}|$ above this range represent the minimum effects that can be detected with adequate statistical power.

For H4a-b, similar G*Power sensitivity analysis parameters used for H3a-l were applied to determine the smallest detectable incremental effect sizes (ΔR^2) for the sequential linear regression models. For H4a, both the predictor of interest and the total number of predictors were identical to those in H3a-f, yielding a minimal detectable incremental effect size of $f^2 = 0.089$. For H4b, three predictors (x-flips, opposite displacement, and time to peak velocity) were entered at the final step of the model, resulting in 11 total predictors. The minimum detectable incremental effect size was $f^2 = 0.127$. Expressed in terms of explained variance, these values correspond to detectable increases in ΔR^2 of approximately 6–8% for H4a and 8–11% for H4b, when assuming plausible full-model R^2 values between 0.10 and 0.40.

Because testing H4c requires comparing non-nested models using information criteria, standard a priori power or sensitivity analyses based on null-hypothesis significance testing are not applicable. Model comparison will therefore rely on relative fit indices (i.e., Akaike Information Criterion [AIC] and Bayesian Information Criterion [BIC]) rather than formal significance testing.

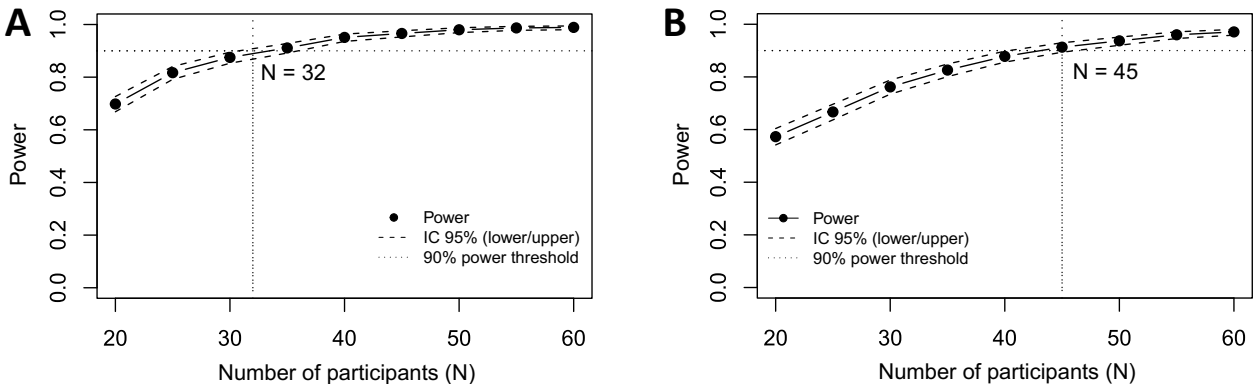

Figure 1. Simulation-based power curves for the interaction between Movement (approach vs. avoidance) and Stimulus Type (experimental vs. neutral) for physical activity **(A)** and sedentary stimuli **(B)**. Note: IC 95% = 95% confidence interval.

Study Procedure

Participants perform the KAAT on the Kinarm Endpoint robotic platform and the manikin task on a laptop using Inquisit 6 software (Millisecond Software, 2015). Task order is randomized, with a five-minute rest period between tasks. Questionnaires are administered after both tasks are completed and collect information on age, sex, gender, weight, height, explicit attitudes towards physical activity, intention to be physically active, physical effort valuation, usual level of physical activity, and automaticity towards physical activity (Figure 2). One attention-check question is included in the questionnaires ("Please answer "2" to this question that allows us to verify that you actually read the questions"). At the end of the session, participants are provided with an ActiGraph wGT3X-BT accelerometer to record their physical activity over the subsequent 7 days.

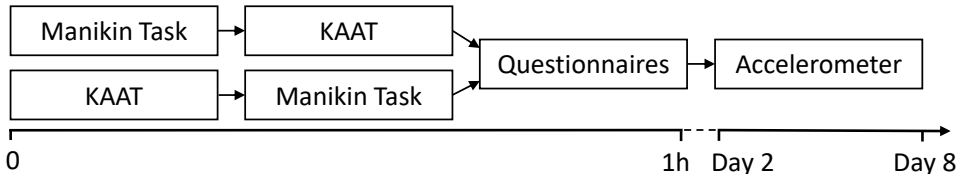

Figure 2. Study timeline. Note: KAAT = Kinarm approach-avoidance task

Approach-Avoidance Tasks

Kinarm Approach-Avoidance Task (KAAT)

The Kinarm Endpoint robotic platform is equipped with a 2-dimensional augmented reality display and provides high-resolution kinematic data. In the KAAT, participants stand while holding the robot's handles, with each hand represented on the display by a white cursor (Figure 4A).

The KAAT begins with on-screen instructions illustrating the upcoming condition. To initiate each block of trials, participants use either hand to place the white cursor into a green circle located at the centre of the workspace. For each trial, a blue circle then appears either to the left or the right side of the display with equal probability. Once the participant places the white cursor of the appropriate hand (i.e., blue circle in the left side requires the left-hand cursor) into the blue circle and holds it there for 500 to 750 ms, a stimulus target appears at one of twelve possible locations positioned 10 cm from the start target and separated by 30 degrees increments (Figure

4B). Simultaneously, a second target consisting of an open white circle appears at the opposite location.

Participants are instructed that bringing the white cursor to the stimulus target corresponds to an "approach" response, whereas bringing the cursor to the opposite target corresponds to an " avoid" response. If neither target is reached within 3000 ms, a "Too Slow!" message is displayed for 750 ms and the trial ends. If the participant incorrectly approaches or avoids a target, an "Incorrect!" text is displayed for 750 ms before the trial ends (Figure 4C) (34). A demonstration video of the KAAT is available in GitHub (45).

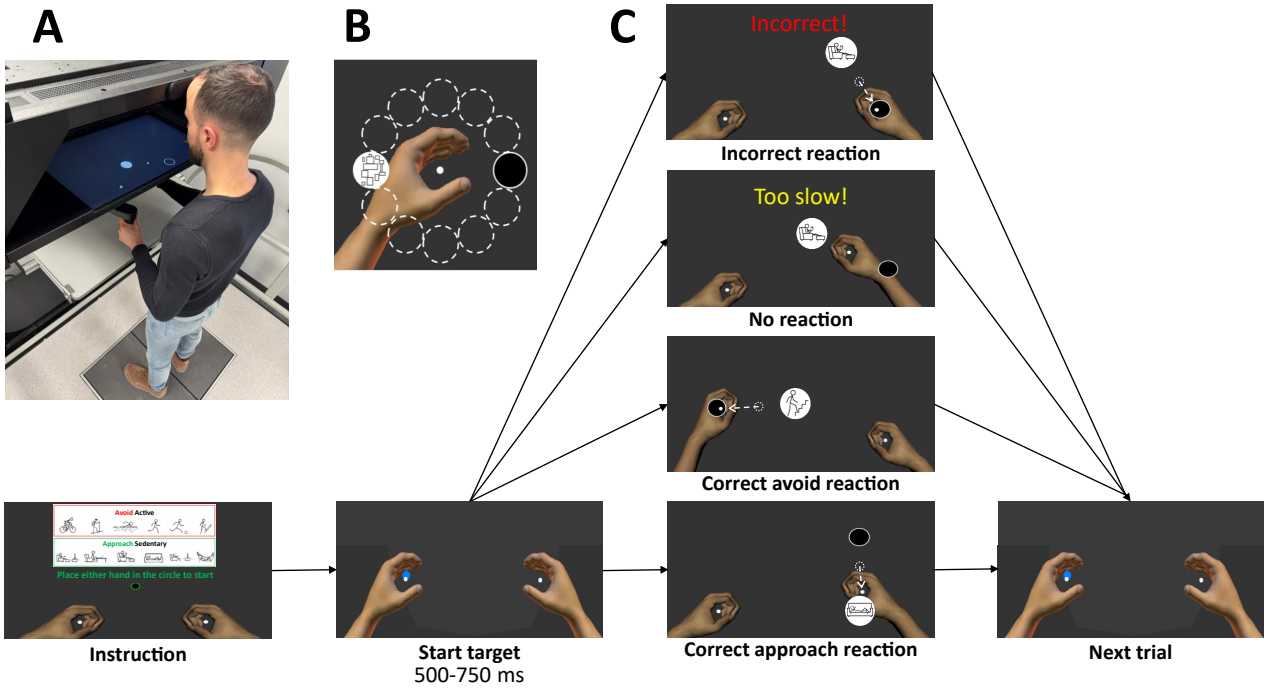

Figure 4. Kinarm Approach-Avoidance Task (KAAT). (A) Experimental setup. **(B)** Example trial in which a neutral stimulus appears in the west position and the open circle appears at the opposite location. Dashed circles illustrate the possible stimulus locations for visualisation purposes only; they are not visible to participants during the task. **(C)** Example trial in which the participant is instructed to approach sedentary stimuli and avoid physical activity stimuli. Hands are shown for illustrative purposes only and are not visible to participants during the actual task.

Manikin Task

The manikin task was implemented in Inquisit 6 software (11,17,27). Participants performed the task in a standing position with the laptop in front of them, with one index finger on the "U" key and the other on the "N" key (Figure 3A). Pressing the "U" key moves the avatar upward, whereas pressing the "N" key moves the avatar downward. When the avatar appears below the stimulus, the "U" key is associated with an approach movement, and the "N" key is associated with an avoidance movement. The opposite mapping applies when the avatar appears above the stimulus.

The manikin task begins with on-screen textual instructions indicating the upcoming condition. Each trial then starts with a fixation cross presented at the centre of the screen for a

random time ranging from 500 to 750 ms (Figure 3B). An avatar subsequently appears at either the top or the bottom of the screen for 1000 ms, after which a stimulus is presented at the centre of the screen. Pressing the correct key triggers the onset of the next trial. If a participant presses the incorrect key (e.g., pressing "U" when "N" is required), an "ERROR" message is displayed for 800 ms and the trial ends. If no key is pressed within seven seconds, a "TOO SLOW" message is displayed for 800 ms before the trial ends. A demonstration video of the manikin task is available in GitHub (45).

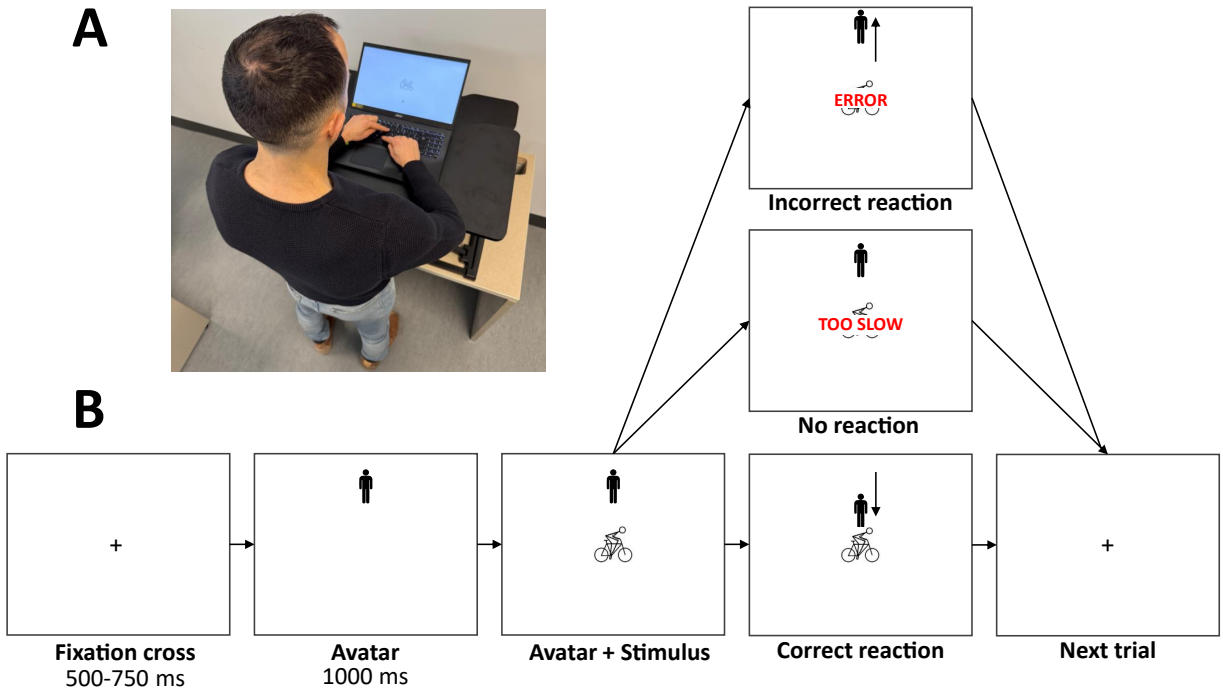

Figure 3. Manikin Task. (A) Experimental setup for the manikin task. **(B)** Example trial of the manikin task in which the participant is instructed to approach physical activity stimuli.

Experimental Conditions

In both the KAAT and the manikin task, task performance was recorded in two experimental conditions and two neutral conditions (Figure 5). Neutral trials were used to compute relative bias (see *Raw and Relative Automatic Approach-Avoidance Bias* section) (11).

In one experimental condition, participants are instructed to approach pictograms depicting physical activity and to avoid pictograms depicting sedentary behaviour. In the other experimental condition, the mapping is reversed: participants avoid physical activity stimuli and approach sedentary stimuli. Automatic tendencies are derived from the time required to press the key in response to a type of stimulus (i.e., physical activity vs. sedentary stimulus) in the manikin task or to reach the target circle with the handle in the KAAT.

In the neutral conditions, experimental pictograms are replaced by stimuli with geometric shapes (circles or squares) matched in number and size of information to three physical activity pictograms (swimming, hiking, cycling) and three sedentary pictograms (couch, hammock, reading) (11,17,27,34). In one neutral condition, participants are instructed to approach circles and avoid squares. In the other condition, the mapping is reversed.

One neutral block is administered at the beginning of the task and the other at the end. The two experimental blocks are completed between the neutral blocks. Each block includes 48 trials, resulting in a total of 192 trials. Approach-avoidance response mapping is counterbalanced across participants separately for neutral (circles vs. squares) and experimental blocks (physical activity vs. sedentary behaviour). Within each block, pictogram presentation order is randomized across both type of stimuli.

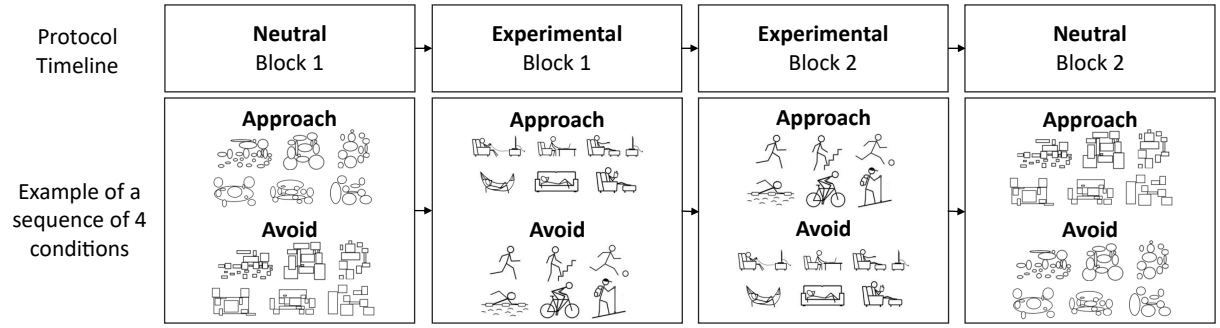

Figure 5. Timeline and stimuli used in the manikin task and the Kinarm Approach-Avoidance Task (KAAT). Approach-avoidance mapping is counterbalanced for participants separately for neutral (circles vs. squares) and experimental blocks (physical activity vs. sedentary behaviours). Within each block, pictogram presentation order is randomized across both type of stimuli.

Reaction Times

Incorrect responses, responses faster than 150 ms, and responses slower than 1500 ms are excluded from the analyses. The lower threshold was selected as a trade-off between the retention of valid observations and the exclusion of anticipatory responses (46). The upper threshold was chosen based on prior work indicating this cut-off yields reliable estimates and maximizes sensitivity (12). Familiarisation trials (the first 15 trials of the study and the first 3 trials of each subsequent condition) will be removed from the analyses (17).

For the manikin task, reaction time is defined as the interval between stimulus onset and the key press. For the KAAT, reaction time is defined as the interval between peripheral target onset and movement initiation, which is identified kinematically by first detecting the time point at which acceleration exceeds 1000 mm/s² and then tracing backward to the last time point at which acceleration was above 200 mm/s². This procedure has been shown to provide a reliable and accurate estimate of movement initiation based on kinematic data (47).

Error Rate

Error rate is defined as the proportion of incorrect or time-out responses relative to the total number of trials, following the same exclusion criteria applied to reaction times. Higher values indicate poorer response accuracy.

Elapsed Time and Task Order

Elapsed time is defined as the duration between task onset (manikin or KAAT) and the response in a given trial. Task order (manikin first vs. KAAT first) will also be included as a

categorical variable. These variables account for potential temporal and order-related effects, including fatigue and learning.

Raw and Relative Automatic Approach-Avoidance Bias

For construct and convergent validity, the raw approach-avoidance bias towards physical activity and sedentary behavior is estimated from the simple effect of Movement (approach vs. avoidance) in the linear mixed-effects models, with Stimulus Type centred on the experimental stimuli. The relative bias, which accounts for a potential generic approach-avoidance tendency that may vary across participants (11,48), is estimated from the interaction effect between Movement and Stimulus Type. Algebraically, under effect coding, this interaction term corresponds to:

$$\begin{aligned}\beta_{Movement \times Stimulus\ Type} &= \left(E[RT_{\text{Avoid Experimental}}] - E[RT_{\text{Approach Experimental}}]\right) \\ &- \left(E[RT_{\text{Avoid Neutral}}] - E[RT_{\text{Approach Neutral}}]\right)\end{aligned}$$

where $E[RT]$ denotes the model-implied expected reaction time. Thus, β is the trial-level analogue of the traditional subtraction computed from aggregated median reaction times.

Although mixed-effects models can produce participant-specific estimates by combining trial-level data, these estimates are partially adjusted (or "shrunk") towards the group average. While this shrinkage improves the stability of the estimates, it can attenuate (i.e., weaken) observed associations when these model-based estimates are subsequently used as predictors in separate analyses (49,50). To avoid this issue, the relative bias towards physical activity used for the assessment of concurrent and predictive validity is calculated directly at the participant level using median reaction times:

$$\begin{aligned}\text{Relative Bias}_{\text{Physical Activity}} &= \left(median\ RT_{\text{Avoid Physical Activity}} - median\ RT_{\text{Approach Physical Activity}}\right) \\ &- \left(median\ RT_{\text{Avoid Neutral}} - median\ RT_{\text{Approach Neutral}}\right)\end{aligned}$$

Positive values indicate a tendency to approach rather than avoid physical activity.

Kinematic Variables

In addition to reaction time, three kinematic outcomes are computed to capture complementary aspects of movement dynamics: a derivative measure, a complexity measure, and a measure of deviation (Figure 6).

The derivative measure, time to peak velocity, reflects reluctance to commit to action. It is defined as the time elapsed from movement onset to the time point of maximum movement velocity (51,52). Movement onset is defined using an algorithm based on acceleration. From stimulus appearance, the algorithm first identifies the time point at which maximum hand acceleration reaches 1000 mm/s². It then traces backward in time to determine when acceleration first reaches 200 mm/s², which is defined as movement onset. Time to peak velocity is computed as the mean time to peak velocity across correct trials, with longer times to peak velocity indicating delayed motor commitment.

The complexity measure, x-flips, reflects wavering during movement execution. It is defined as the number of directional changes along with the x-axis (52,53), which are computed

as the number of sign changes in velocity after excluding small variations around zero. This variable is computed as the mean number of x-flips across correct trials, with higher number of x-flips indicating greater decision difficulty (54).

The deviation measure, opposite displacement, reflects response competition by capturing the extent to which the movement initially unfolds in the incorrect direction. It is defined as the mean displacement in the direction opposite to the correct response, calculated on correct trials. This variable is computed as the mean opposite displacements across correct trials, with higher distance indicating greater response competition.

To control for a potential generic approach or avoidance bias (11), the participant's mean value in the approach neutral stimulus condition is subtracted from the values in each experimental condition for each kinematic outcome.

These kinematic outcomes reflect approach-avoidance conflicts (55), as altered movement dynamics suggest that competing approach and avoidance tendencies were concurrently activated, resulting in a conflict between task instructions and underlying tendencies during movement execution.

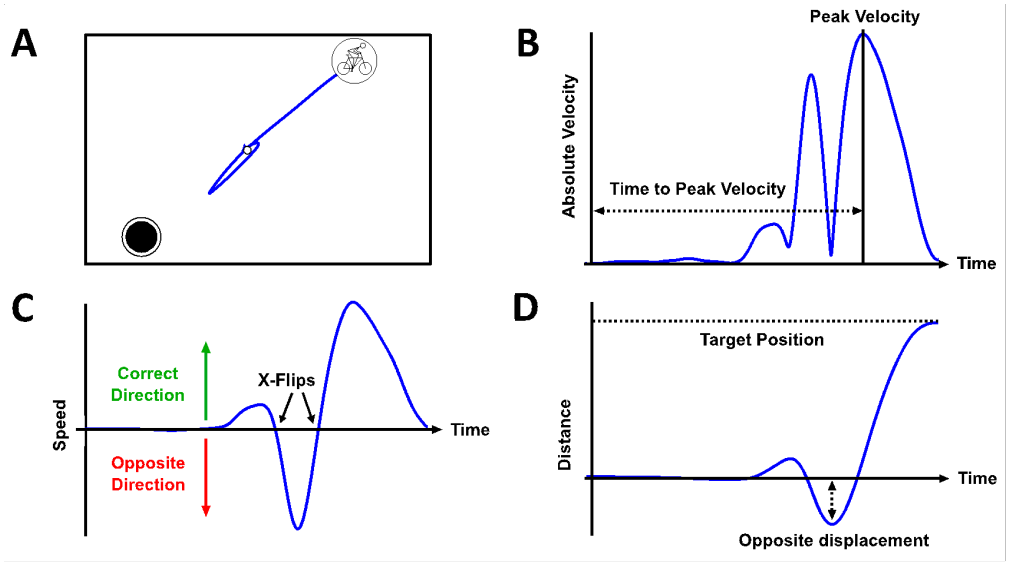

Figure 6. Kinematic Variables. (A) Example trial from the Kinarm approach-avoidance task (KAAT) with corresponding kinematic variables shown in panel **(B)** time to peak velocity, **(C)** number of x-flips, and **(D)** opposite displacement.

Reliability of Participant-Level Relative Bias

To evaluate the reliability of the relative approach-avoidance bias at the participant level, split-half reliability is estimated using a resampling-based procedure combined with intraclass correlation coefficients (ICC).

For each participant and for each condition (physical activity and sedentary behavior), trials are randomly divided into two independent halves using stratified sampling to ensure that each half contains an equal number of observations from each combination of Movement (approach vs. avoidance) and Stimulus Type (experimental vs. neutral). Median reaction times are then computed separately for each half and used to calculate the relative approach-avoidance bias using the same difference-of-differences formula described above, yielding two independent bias estimates per participant.

To reduce dependency on any single arbitrary split and to obtain a more stable estimate of internal consistency, this procedure is repeated 500 times using repeated random stratified splits. For each iteration, agreement between the two bias estimates is quantified using a two-way random-effects ICC with absolute agreement (ICC[2,1]) (56,57). ICC values are then averaged across iterations, and 95% confidence intervals will be reported to provide a robust estimate of reliability.

Because split-half reliability reflects the consistency of a half-length measure, coefficients are adjusted using the Spearman-Brown prophecy formula to estimate reliability of the full-length task (58). Higher ICC values indicate greater within-participant stability and thus greater reliability of the bias measure. Following conventional guidelines, ICC values < .40 will be interpreted as poor, .40–.59 as fair, .60–.74 as good, and ≥ .75 as excellent reliability (57).

Device-Based Level of Physical Activity

Accelerometer Measurement

Device-based physical activity is assessed using a 3-axis ActiGraph wGT3X-BT accelerometer sampling at 30 Hz. The accelerometer is worn on the right hip using a belt clip and elastic waistband. Participants are instructed to wear the accelerometer for seven consecutive days during all waking hours, for at least 10 hours per day, except during aquatic activities (e.g., bathing, swimming). A monitoring week is considered valid if it includes at least four valid days, including one weekend day (59–61).

Wear and non-wear time are determined using the algorithm implemented in ActiLife v6.15.0, which defines non-wear time as ≥ 60 consecutive minutes of 0 counts per minute (cpm), permitting brief interruptions of non-zero counts within that window (59). Participants also record their sleep periods and non-wear time in a daily log to verify the algorithm's classification.

Previous work derived ActiGraph accelerometer cut-points for classifying physical activity intensity using receiver operating characteristic (ROC) curve analyses, selecting thresholds that maximized the sum of sensitivity and specificity for moderate (≥ 3 METs) and vigorous (≥ 6 METs) intensity levels (62). Accordingly, vector magnitude counts are aggregated into 60-second epochs and classified as sedentary behaviour (< 200 cpm) (63,64), light intensity (200-4977 cpm), moderate intensity (4978-6226 cpm), and vigorous intensity (> 6226 cpm) (62).

Total Physical Activity

Light-intensity physical activity is associated with measurable health benefits (65) and has been shown to attenuate the association between sedentary time and mortality independently of moderate-to-vigorous physical activity (66). Moreover, recent evidence indicates that total physical activity volume across all intensities is more strongly associated with mortality reduction than moderate-to-vigorous physical activity alone (67). Consistent with this evidence, recent large accelerometer-based cohort studies have adopted total activity as a primary outcome (66,68).

Accordingly, in the present study, time spent in each category is averaged across valid days (minutes/day) and weighted by the assigned metabolic equivalent of task (MET) assigned: 2 METs for light physical activity, 4.5 METs for moderate physical activity, and 8 METs for vigorous physical activity (69). The weighted average values are then multiplied by seven to obtain a weekly total expressed in MET-min/week, reflecting weekly physical activity volume adjusted for intensity. In addition, device-based moderate-to-vigorous physical activity in min/week is calculated by averaging daily time spent in moderate and vigorous intensity activity and multiplying this value by seven.

Questionnaires

Demographic and Anthropometric Variables

Sex (male, female), age (years), height (cm), weight (kg), and handedness were self-reported.

Intention to Be Physically Active

Intention to be physically active is assessed with the item: "Over the next 7 days, I intend to do at least 150 minutes of moderate-intensity physical activity; or at least 75 minutes of vigorous intensity physical activity; or an equivalent combination of moderate- and vigorous-intensity physical activity". Responses are provided on a 7-point Likert scale ranging from 1 ("strongly disagree") to 7 ("strongly agree") (11).

Explicit Affective Attitudes Towards Physical Activity

Explicit attitudes towards physical activity are assessed using two bipolar semantic differential items introduced by the stem: "For me, to participate in regular physical activity is …" (11,70). Responses are provided on a 7-point Likert scale for the adjective pairs "unpleasant" (1) to "pleasant" (7) and "unenjoyable" (1) to "enjoyable" (7). Scores are computed as the mean of the two items, with higher values indicating more positive affective attitudes towards physical activity.

Automaticity Towards Physical Activity

Automaticity towards physical activity is assessed using the automaticity subscale of the Self-Reported Habit Index (71,72). Participants first read the following stem "Overall, the decision to engage in physical activity is something…". They then indicate their agreement with four statements: "I do automatically", "I do without having to consciously remember", "I do without thinking", and "I start doing before I realize I'm doing it" (a = .94). Responses are provided on a 7-point Likert scale ranging from 1 ("Strongly disagree") to 7 ("Strongly agree"). Scores are computed as the mean of the four items, with higher scores indicating that engaging in physical activity is perceived as more automatic and habitual, whereas lower scores reflect more deliberate processes.

Valuation of Physical Effort

Valuation of physical effort is assessed using the Physical Effort Scale (73,74), an 8-item questionnaire with two dimensions: approach and avoidance of physical effort. Items are rated on a 5-point Likert scale ranging from 1 ("I completely disagree") to 5 ("I completely agree") and include statements such as "I usually like activities that require physical effort" and "I tend to avoid situations in which I have to exert physical effort". Scores are computed as the mean of the eight items, with higher scores on the approach dimension reflecting greater positive valuation of physical effort, whereas higher scores on the avoidance dimension reflect stronger tendencies to avoid effort.

Descriptive Analyses

To characterise the sample, we will report means and standards deviations for age, height, weight, moderate-to-vigorous physical activity, total physical activity, intentions, explicit attitudes, automaticity, and the valuation of physical effort. Sex and handedness will be

summarised using counts and percentages. Descriptive statistics will also be calculated for reaction times, number of errors, and kinematic variables, stratified by stimulus type and each movement type.

Statistical Analysis

Construct Validity (H1): Linear Mixed-Effects Models

To test H1, linear mixed-effects models will be fitted in the R software environment (75) using the *lme4* (76) and *lmerTest* packages (77). The eight sub-hypotheses (H1a-H1h; Table 1) will be tested using the following model specification (R formula syntax), estimated separately for physical activity and sedentary behaviour:

$$\begin{aligned} Reaction\ Time \sim Movement\ *\ Stimulus\ Type\ *\ Task + Age + Sex +\ Height * Weight \\ + \text{Handedness} + \text{Error Rate} + \text{Elapsed Time} + \text{Task Order} + (1 + \text{Task} \\ + \text{Movement} * \text{Stimulus Type} \mid \text{Subject}) + (1 \mid \text{Pictogram}) \end{aligned}$$

Categorical predictors will be deviation-coded (−0.5/+0.5), such that main effects represent effects averaged across the levels of the other interacting factors. Continuous covariates will be standardized (z-scored) prior to analysis to facilitate model convergence and interpretation. Random effects will include intercepts for Subject and Pictogram. For participants, random slopes will be specified for Task and for the Movement × Stimulus Type effect, allowing individual differences in overall task effects and approach-avoidance patterns.

In the case of non-convergence or singular fit, model simplification will follow a predefined sequence. First, correlations between random slopes and intercepts will be removed. Second, random slopes associated with higher-order interaction terms will be sequentially removed. As a final step, non-essential covariates may be removed, while all primary fixed effects and their interactions will be retained. This predefined simplification strategy aims to balance model complexity with numerical stability and statistical power (42).

For the hypotheses testing internal structure focusing on the KAAT, H1a and H1b (KAAT raw bias) will be tested using the simple effect of Movement (approach vs. avoidance), evaluated at the level of Task = KAAT and with Stimulus Type evaluated at the level of experimental stimuli (physical activity for H1a and sedentary behaviour for H1b). This parameter reflects the raw approach-avoidance bias for physical activity (H1a) and sedentary stimuli (H1b). H1c and H1d (KAAT relative bias) will be tested using the Movement × Stimulus Type interaction, evaluated at Task = KAAT (see section *Raw and Relative Automatic Approach-Avoidance Bias*).

For the sensitivity hypotheses comparing tasks, H1e and H1f (difference in raw bias between tasks) will be tested using the Movement × Task interaction, evaluated at the level of experimental stimuli. H1g and H1h (difference in relative bias between tasks) will be tested using the Movement × Stimulus Type × Task interaction, which evaluates whether the neutral-corrected bias differs between the KAAT and the manikin task.

Models will be estimated using restricted maximum likelihood (REML). Statistical assumptions will be evaluated using residual diagnostics (Q-Q plots and residual-versus-fitted plots). If substantial deviations from normality or homoscedasticity are observed, sensitivity analyses will be conducted using log-transformed reaction times to assess the robustness of the findings. Statistical significance will be evaluated at $\alpha = .05$ using Satterthwaite-approximated degrees of freedom, and effect sizes with 95% confidence intervals will be reported.

Convergent Validity (H2): Bivariate Correlations

To assess convergent validity (H2), Pearson correlations will be computed between KAAT and manikin indices of raw approach-avoidance bias (H2a and H2b) and relative approach-avoidance bias (H2c and H2d), separately for physical activity and for sedentary stimuli. Correlation coefficients will be reported alongside 95% confidence intervals and corresponding p-values. The significance level will be set at $\alpha = .05$. Given prior evidence of limited convergence and substantial variability between approach-avoidance tasks (13,35), results will be interpreted with primary emphasis on effect size estimates and their precision rather than statistical significance alone.

Concurrent Validity (H3): Multiple Linear Regressions

Concurrent validity will be assessed by examining whether KAAT-derived relative approach-avoidance bias is associated with theoretically related behavioral and motivational indicators measured at the same time point (H3a-H3f), and whether these associations are stronger than those observed for the manikin task (H3g-H3l).

For H3a-H3f, separate multiple regressions will be estimated for each outcome variable: device-based total physical activity, device-based moderate-to-vigorous physical activity, explicit affective attitude towards physical activity, intention to be physically active, automaticity towards physical activity, and valuation of physical effort. Each model will include age, sex, height, weight, handedness, error rate, and elapsed time as covariates, with KAAT relative bias as the predictor of interest. The association between KAAT relative bias and each outcome will be evaluated using the standardized regression coefficient (β), its 95% confidence interval, and corresponding *p*-value.

For H3g-H3l, separate multiple regressions will include the same covariates, along with both KAAT relative bias and manikin relative bias. Including both biases in the same model allows estimation of their unique (partial) associations with each outcome while accounting for shared variance. Standardized regression coefficients and 95% confidence intervals will be reported for each task-specific bias. To formally test whether the KAAT relative bias is more strongly associated with each outcome than manikin bias, planned contrasts comparing regression coefficients ($\beta_{KAAT} - \beta_{manikin}$) will be conducted within each model. A significant positive contrast will indicate that the KAAT-derived bias explains more variance in the outcome than the manikin-derived bias.

All continuous covariates and predictors will be standardized prior to analysis. Multicollinearity will be assessed using variance inflation factors (VIF), with values above 5 considered indicative of potentially problematic collinearity. If detected, this will be transparently reported and considered in the interpretation of regression estimates rather than addressed through automatic variable exclusion (78).

To control for multiple comparisons within H3a–H3f and H3g–H3l (79), the six hypothesis tests in each set will be treated as a single family. *P*-values will be adjusted using the Benjamini-Hochberg False Discovery Rate procedure ($q = .05$) (80,81). Device-based total physical activity will be specified a priori as the primary outcome within each family and will be evaluated without correction.

Predictive Validity (H4): Sequential Regressions and Model Comparison

Predictive validity will be evaluated by examining whether KAAT-derived measures explain variance in device-based total physical activity (H4a and H4b) and whether their predictive performance exceeds that of the manikin task (H4c).

For H4a and H4b, a multiple linear regression model will be estimated in three sequential steps. *In Step 1*, age, sex, height, weight, handedness, error rate, and elapsed time will be entered as covariates. In *Step 2,* KAAT relative bias for physical activity will be added. *In Step 3*: kinematic variables derived from the KAAT (time to peak velocity, number of x-flips, opposite displacement) will be added.

H4a will be evaluated by examining the change in explained variance (ΔR^2) when KAAT relative bias is added at Step 2. A statistically significant increase in ΔR^2 at this step will indicate that KAAT relative bias explains variance in device-based total physical activity beyond covariates. H4b will be evaluated by examining ΔR^2 when the kinematic variables are added at Step 3. A statistically significant increase in ΔR^2 at this step will indicate incremental predictive validity of the kinematic variables beyond relative bias. Changes in explained variance will be tested using F-tests for nested models. Standardized regression coefficients (β), 95% confidence intervals, and ΔR^2 values will be reported. Multicollinearity will be assessed as described for H3.

To test H4c, a separate regression including the same covariates as in Step 2 will be estimated, replacing the KAAT-derived relative bias with the manikin-derived relative bias. The predictive performance of the KAAT-based models (Step 2) and the manikin-based model will then be compared. Because these models are not nested, comparisons will rely on adjusted R^2 and information criteria (AIC and BIC) (82,83) rather than F-tests. Lower AIC and BIC values indicate better model fit. Differences in AIC (ΔAIC) will be interpreted using established guidelines (84), with values ≤ 2 indicating similar support for both models, values between 4 and 7 indicating less support for the model with the higher AIC, and values > 10 indicating no support for the model with the higher AIC. Differences in BIC (ΔBIC) will be interpreted using established guidelines (85), with values < 2 indicating negligible evidence, values between 2 and 6 indicating positive evidence, values between 6 and 10 indicating strong evidence, and values > 10 indicating very strong evidence against the model with the higher BIC. In addition, Akaike weights will be computed to quantify the relative support for each model, expressed as a probability-like measure of model plausibility (84,86).

Data & Code Accessibility

Data, R scripts, and material used for this study are publicly available (87).

Ethical Statement

The study was approved by University of Ottawa's Research Ethics Boards (H-01-25-11075) and informed consent was collected in accordance with the Declaration of Helsinki.

Use of Artificial Intelligence (AI) and AI-assisted technologies

ChatGPT (OpenAI) and DeepL were used to refine the language, improve readability, and correct grammar and spelling errors of this manuscript.

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Competing Interests

The authors declare no financial conflicts of interest related to the content of this article.

Authors' Contributions

Based on the Contributor Roles Taxonomy (CRediT) (88) individual author contributions to this work will be as follows: François Jabouille: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Visualization, Writing – Original Draft, Writing – Review and Editing, Supervision; Timothée Dumas: Data Curation, Visualization, Writing – Review and Editing; Jérémy Fortier: Investigation; Michaël Patrick Robichaud: Investigation; Brian Nhat Thien Nguyen: Investigation; Frédérik Jolicoeur: Investigation; Silvio Maltagliati: Writing – Review and Editing; Boris Cheval: Writing – Review and Editing; Kayne Park: Methodology, Data Curation, Visualization, Writing – Review and Editing; Matthieu Boisgontier: Conceptualization, Methodology, Formal Analysis, Visualization, Writing – Original Draft, Writing – Review and Editing, Supervision, Project Administration, Funding Acquisition.

REFERENCES

1. Hofmann W, Friese M, Wiers RW. 2008. Impulsive versus reflective influences on health behavior: a theoretical framework and empirical review. *Health Psychol Rev*. 2(2):111–137. https://doi.org/10.1080/17437190802617668
2. Sheeran P, Gollwitzer PM, Bargh JA. 2013. Nonconscious processes and health. *Health Psychol.* 32(5):460–473. https://doi.org/10.1037/a0029203
3. Cheval B, Sarrazin P, Pelletier L. 2014. Impulsive approach tendencies towards physical activity and sedentary behaviors, but not reflective intentions, prospectively predict non-exercise activity thermogenesis. *PLoS One.* 9(12):e115238. https://doi.org/10.1371/journal.pone.0115238
4. Cheval B, Tipura E, Burra N, Frossard J, Chanal J, Orsholits D, Radel R, Boisgontier MP. 2018. Avoiding sedentary behaviors requires more cortical resources than avoiding physical activity: an EEG study. *Neuropsychologia.* 119:68–80. https://doi.org/10.1016/j.neuropsychologia.2018.07.029
5. Cheval B, Sarrazin P, Isoard-Gautheur S, Radel R, Friese M. 2015. Reflective and impulsive processes explain (in)effectiveness of messages promoting physical activity: a randomized controlled trial. *Health Psychol.* 34(1):10–19. https://doi.org/10.1037/hea0000102
6. Conroy DE, Hyde AL, Doerksen SE, Ribeiro NF. 2010. Implicit attitudes and explicit motivation prospectively predict physical activity. *Ann. Behav. Med.* 39(2):112–118. https://doi.org/10.1007/s12160-010-9161-0
7. Wang Y, Zhang Z, Mao Z. 2024. You don't know why you (don't) exercise: the relationship between automatic processes and physical activity (or sedentary behavior): a meta-analysis. *Scand. J. Psychol.* 65(3):501–510. https://doi.org/10.1111/sjop.12991
8. Evans JSBT, Stanovich KE. 2013. Dual-process theories of higher cognition: advancing the debate. *Perspect. Psychol. Sci.* 8(3):223–241. https://doi.org/10.1177/174569161246068
9. Chen M, Bargh JA. 1999. Consequences of automatic evaluation: immediate behavioral predispositions to approach or avoid the stimulus. *Pers. Soc. Psychol. Bull*. 25(2):215–224. https://doi.org/10.1177/0146167299025002007
10. Rinck M, Becker ES. 2007. Approach and avoidance in fear of spiders. *J. Behav. Ther. Exp. Psychiatry*. 38(2):105–120. https://doi.org/10.1016/j.jbtep.2006.10.001
11. Farajzadeh A, Goubran M, Beehler A, Cherkaoui N, Morrison P, De Chanaleilles M, Maltagliati S, Boris C, Miller MW, Sheehy L, et al. 2023. Automatic approach-avoidance tendency toward physical activity, sedentary, and neutral stimuli as a function of age, explicit affective attitude, and intention to be active. *Peer Community J.* 3:e21. https://doi.org/10.24072/pcjournal.246
12. Krieglmeyer R, Deutsch R. 2010. Comparing measures of approach–avoidance behaviour: the manikin task vs. two versions of the joystick task. *Cogn. Emot*. 24(5):810–828. https://doi.org/10.1080/02699930903047298

13. Wittekind CE, Blechert J, Schiebel T, Lender A, Kahveci S, Kühn S. 2021. Comparison of different response devices to assess behavioral tendencies towards chocolate in the approach-avoidance task. *Appetite*. 165:105294. https://doi.org/10.1016/j.appet.2021.105294
14. Braunstein LM, Kuerbis A, Ochsner K, Morgenstern J. 2016. Implicit alcohol approach and avoidance tendencies predict future drinking in problem drinkers. *Alcohol. Clin. Exp. Res*. 40(9):1945–1952. https://doi.org/10.1111/acer.13151
15. Evans TC, Taylor CT, Britton JC. 2021. Characterizing the time course of automatic action tendencies to affective facial expressions and its dysregulation in social anxiety disorder. *J. Anxiety Disord*. 78:102363. https://doi.org/10.1016/j.janxdis.2021.102363
16. Deuter CE, Smit J, Kaczmarczyk M, Wingenfeld K, Otte C, Kuehl LK. 2021. Approach-avoidance tendencies in depression and childhood trauma: no effect of noradrenergic stimulation. *Compr. Psychoneuroendocrinol.* 8:100077. https://doi.org/10.1016/j.cpnec.2021.100077
17. Farajzadeh A, Jabouille F, Benoit N, Bezeau O, Bourgie T, Gerro B, Ouimet J, Boisgontier MP. 2024. Apathy, Intentions, Explicit Attitudes, and Approach-Avoidance Tendencies in Physical Activity Behavior. *Commun. Kinesiol.* 1(6). https://doi.org/10.51224/cik.2024.68
18. Cheval B, Sarrazin P, Isoard-Gautheur S, Radel R, Friese M. 2016. How impulsivity shapes the interplay of impulsive and reflective processes involved in objective physical activity. *Pers. Individ. Dif.* 96:132–137. https://doi.org/10.1016/j.paid.2016.02.067
19. Daou M, Lohse K, Miller M. 2017. To take the stairs or not to take the stairs? Employing the reflective–impulsive model to predict spontaneous physical activity. *Sports.* 5(4):75. https://doi.org/10.3390/sports5040075
20. Oliver S, Kemps E. 2018. Motivational and implicit processes contribute to incidental physical activity. *Br. J. Health Psychol.* 23(4):820–842. https://doi.org/10.1111/bjhp.12318
21. Hannan TE, Moffitt RL, Neumann DL, Kemps E. 2019. Implicit approach-avoidance associations predict leisure-time exercise independently of explicit exercise motivation. *Sport. Exerc. Perform. Psychol*. 8(2):210–222. https://doi.org/10.1037/spy0000145
22. Zenko Z, Ekkekakis P. 2019. Internal consistency and validity of measures of automatic exercise associations. *Psychol. Sport Exerc.* 43:4–15. https://doi.org/10.1016/j.psychsport.2018.12.005
23. Berry TR. 2020. You do (not?) have to go to the gym: effects of negations in exercise messages. *Sport. Exerc. Perform. Psychol.* 9(3):437–449. https://doi.org/10.1037/spy0000185
24. Moffitt RL, Kemps E, Hannan TE, Neumann DL, Stopar SP, Anderson CJ. 2020. Implicit approach biases for physically active lifestyle cues. *Int. J. Sport Exerc. Psychol*. 18(6):833–849. https://doi.org/10.1080/1612197X.2019.1581829
25. Wang YB, Sun XT, Mao ZX. 2023. Is approaching sedentary behaviour or physical activity a reward? An EEG study. *Behav. Brain Res*. 445:114364. https://doi.org/10.1016/j.bbr.2023.114364
26. Fessler L, Tessitore E, Craviari C, Sarrazin P, Meyer P, Luthy C, et al. 2024. Motivational and emotional correlates of physical activity and sedentary behavior after cardiac rehabilitation: an

observational study. *BMC Sports Sci. Med. Rehabil*. 16(1):209. https://doi.org/10.1186/s13102-024-00997-0

27. Goubran M, Zammar C, Tellez Alvarez S, Héran É, Proulx S, Bilodeau M, Boisgontier MP. 2025. Approach-avoidance tendencies influence the relationship between fear of movement and physical activity in osteoarthritis. *Physiother. Can.* e20250011. https://doi.org/10.3138/ptc-2025-0011
28. Wang C, Zhang G. 2025. Examination of the consistency effect between physical activity levels and approach-avoidance behaviors. *Psychol. Sport Exerc.* 79:102825. https://doi.org/10.1016/j.psychsport.2025.102825
29. Preis MA, Zellerhoff M, Brockmeyer T. 2021. Approach bias modification training to increase physical activity: a pilot randomized controlled trial in healthy volunteers. *J. Health Psychol.* 26(13):2470–2486. https://doi.org/10.1177/1359105320913936
30. Locke SR, Berry TR. 2021. Examining the relationship between exercise-related cognitive errors, exercise schema, and implicit associations. *J. Sport Exerc. Psychol*. 43(4):345–352. https://doi.org/10.1123/jsep.2021-0031
31. Quossi A, Fessler L, Maltagliati S, Gardner B, Miller MW, Sander D, et al. 2025. Can automatic reactions mirror exercise dependence? *Int. J. Sport Exerc. Psychol*. 23(7):1207–1233. https://doi.org/10.1080/1612197X.2024.2387698
32. Cheval B, Boisgontier MP. 2021. The theory of effort minimization in physical activity. *Exerc. Sport Sci. Rev.* 49(3):168–178. https://doi.org/10.1249/JES.0000000000000252
33. Conroy DE, Berry TR. 2017. Automatic affective evaluations of physical activity. *Exerc. Sport Sci. Rev*. 45(4):230–237. https://doi.org/10.1249/JES.0000000000000120
34. Park K, Boisgontier MP. 2025. A novel robotic reaching task to advance the assessment of approach-avoidance tendencies through kinematic analysis. *Peer Community J*. 5:e25. https://doi.org/10.24072/pcjournal.530
35. Fricke K, Alexander N, Jacobsen T, Vogel S. 2023. Comparison of two reaction-time-based and one foraging-based behavioral approach-avoidance tasks in relation to interindividual differences and their reliability. *Sci. Rep.* 13(1):22376. https://doi.org/10.1038/s41598-023-49864-x
36. Westfall J, Kenny DA, Judd CM. 2014. Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli. *J. Exp. Psychol. Gen*. 143(5):2020–2045. https://doi.org/10.1037/xge0000014
37. Brysbaert M, Stevens M. 2018. Power analysis and effect size in mixed effects models: a tutorial. *J. Cogn.* 1(1):9. https://doi.org/10.5334/joc.10
38. Green P, MacLeod CJ. 2016. SIMR: an R package for power analysis of generalized linear mixed models by simulation. *Methods Ecol. Evol*. 7(4):493–498. https://doi.org/10.1111/2041-210X.12504
39. Cheval B, Tipura E, Burra N, Frossard J, Chanal J, Orsholits D, Radel R, Boisgontier MP. 2018. Avoiding sedentary behaviors requires more cortical resources than avoiding physical activity. Zenodo. https://doi.org/10.5281/zenodo.1169140

40. Lohse K, Kozlowski A, Strube MJ. 2023. Model specification in mixed-effects models: a focus on random effects. *Commun. Kinesiol*. 1(5). https://doi.org/10.51224/cik.2023.52
41. Bates D, Mächler M, Bolker B, Walker S. 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Softw*. 67:1-48. https://doi.org/10.18637/jss.v067.i01
42. Matuschek H, Kliegl R, Vasishth S, Baayen H, Bates D. 2017. Balancing Type I error and power in linear mixed models. *J. Mem. Lang.* 94:305–315. https://doi.org/10.1016/j.jml.2017.01.001
43. Morris TP, White IR, Crowther MJ. 2019. Using simulation studies to evaluate statistical methods. *Stat. Med*. 38(11):2074–2102. https://doi.org/10.1002/sim.8086
44. Lakens D. 2022. Sample size justification. Collabra Psychol. 8(1):33267. https://doi.org/10.1525/collabra.33267
45. Jabouille F, Park K, Boisgontier M. 2026. Manikin and KAAT demonstration videos. GitHub. https://github.com/Boisgontier-Lab/RR_Validation_KAAT/tree/main/Manikin_and_KAAT_demonstration_videos
46. Miller J. 2023. Outlier exclusion procedures for reaction time analysis: the cures are generally worse than the disease. *J. Exp. Psychol. Gen*. 152(11):3189–3217. https://doi.org/10.1037/xge0001450
47. Blinch J, Trovinger C, DeWinne CR, De Cellio Martins G, Ifediora CN, Nourollahimoghadam M, Harry JR, Palmer TB. 2023. Tradeoffs of estimating reaction time with absolute and relative thresholds. *Behav. Res. Methods*. 56(5):4695–4715. https://doi.org/10.3758/s13428-023-02211-4
48. Windsor TD, Pearson EL, Butterworth P. 2012. Age group differences and longitudinal changes in approach–avoidance sensitivity: findings from an 8-year longitudinal study. *J. Res. Pers*. 46(6):646–654. https://doi.org/10.1016/j.jrp.2012.07.002
49. Kliegl R. 2010. Experimental effects and individual differences in linear mixed models: estimating the relationship between spatial, object, and attraction effects in visual attention. *Front. Psychol*. 1:238. https://doi.org/10.3389/fpsyg.2010.00238
50. Liu S, Kuppens P, Bringmann L. 2021. On the use of empirical Bayes estimates as measures of individual traits. *Assessment*. 28(3):845–857. https://doi.org/10.1177/1073191119885019
51. Grenke O, Scherbaum S, Schoemann M. 2025. Exploring the synthesis of mouse cursor tracking and drift diffusion modeling in a perceptual decision-making task. *Behav. Res. Methods*. 57(11):297. https://doi.org/10.3758/s13428-025-02805-0
52. Wirth R, Foerster A, Kunde W, Pfister R. 2020. Design choices: empirical recommendations for designing two-dimensional finger-tracking experiments. *Behav. Res. Methods*. 52(6):2394–2416. https://doi.org/10.3758/s13428-020-01409-0
53. Dale R, Roche J, Snyder K, McCall R. 2008. Exploring action dynamics as an index of paired-associate learning. *PLoS One*. 3(3):e1728. https://doi.org/10.1371/journal.pone.0001728
54. Cheng J, González-Vallejo C. 2018. Unpacking decision difficulty: testing action dynamics in intertemporal, gamble, and consumer choices. *Acta Psychol.* 190:199–216. https://doi.org/10.1016/j.actpsy.2018.08.002

853 55. Garcia-Guerrero S, O'Hora D, Zgonnikov A, Scherbaum S. 2023. The action dynamics of
854 approach-avoidance conflict during decision-making. *Q. J. Exp. Psychol*. 76(1):160–179.
855 https://doi.org/10.1177/17470218221087625
856 56. Shrout PE, Fleiss JL. 1979. Intraclass correlations: uses in assessing rater reliability. *Psychol.*
857 *Bull*. 86(2):420–428. https://doi.org/10.1037/0033-2909.86.2.420
858 57. Koo TK, Li MY. 2016. A guideline of selecting and reporting intraclass correlation coefficients
859 for reliability research. *J. Chiropr. Med.* 15(2):155–163.
860 https://doi.org/10.1016/j.jcm.2016.02.012
861 58. de Vet HCW, Mokkink LB, Mosmuller DG, Terwee CB. 2017. Spearman–Brown prophecy
862 formula and Cronbach's alpha: different faces of reliability and opportunities for new
863 applications. *J. Clin. Epidemiol*. 85:45–49. https://doi.org/10.1016/j.jclinepi.2017.01.013
864 59. Migueles JH, Cadenas-Sanchez C, Ekelund U, Delisle Nyström C, Mora-Gonzalez J, Löf M,
865 Labayen I, Ruiz JR, Ortega FB. 2017. Accelerometer data collection and processing criteria to
866 assess physical activity and other outcomes: a systematic review and practical considerations.
867 *Sports Med*. 47(9):1821–1845. https://doi.org/10.1007/s40279-017-0716-0
868 60. Trost SG, McIver KL, Pate RR. 2005. Conducting accelerometer-based activity assessments
869 in field-based research. *Med. Sci. Sports Exerc*. 37(11):S531–S543.
870 https://doi.org/10.1249/01.mss.0000185657.86065.98
871 61. Suorsa K, Leskinen T, Rovio S, Niinikoski H, Pentti J, Nevalainen J, Heinonen OJ, Lagström
872 H, Jula A, Viikari J, et al. 2023. Weekday and weekend physical activity patterns and their
873 correlates among young adults. *Scand. J. Med. Sci. Sports*. 33(12):2573–2584.
874 https://doi.org/10.1111/sms.14475
875 62. Rhudy MB, Dreisbach SB, Moran MD, Ruggiero MJ, Veerabhadrappa P. 2020. Cut points of
876 the Actigraph GT9X for moderate and vigorous intensity physical activity at four different
877 wear locations. *J. Sports Sci.* 38(5):503–510. https://doi.org/10.1080/02640414.2019.1707956
878 63. Heesch KC, Hill RL, Aguilar-Farias N, Van Uffelen JGZ, Pavey T. 2018. Validity of objective
879 methods for measuring sedentary behaviour in older adults: a systematic review. *Int. J. Behav.*
880 *Nutr. Phys. Act*. 15(1):119. https://doi.org/10.1186/s12966-018-0749-2
881 64. Aguilar-Farías N, Brown WJ, Peeters G. 2014. ActiGraph GT3X+ cut-points for identifying
882 sedentary behaviour in older adults in free-living environments. *J. Sci. Med. Sport*. 17(3):293–
883 299. https://doi.org/10.1016/j.jsams.2013.07.002
884 65. Amagasa S, Machida M, Fukushima N, Kikuchi H, Takamiya T, Odagiri Y, et al. 2018. Is
885 objectively measured light-intensity physical activity associated with health outcomes after
886 adjustment for moderate-to-vigorous physical activity in adults? a systematic review. *Int. J.*
887 *Behav. Nutr. Phys. Act.* 15(1):65. https://doi.org/10.1186/s12966-018-0695-z
888 66. Rezende LFM, Ahmadi M, Ferrari G, Del Pozo Cruz B, Lee IM, Ekelund U, et al. 2024.
889 Device-measured sedentary time and intensity-specific physical activity in relation to all-cause
890 and cardiovascular disease mortality: the UK Biobank cohort study. *Int. J. Behav. Nutr. Phys.*
891 *Act.* 21(1):68. https://doi.org/10.1186/s12966-024-01615-5

67. Tarp J, Dalene KE, Fagerland MW, Steene-Johannesen J, Hansen BH, Anderssen SA, Hagströmer M, Dohrn, IM, Dempsey PC, Wijndaele K, et al. 2024. Physical activity volume, intensity, and mortality: harmonized meta-analysis of prospective cohort studies. *Am. J. Prev. Med*. 67(6):887–896. https://doi.org/10.1016/j.amepre.2024.07.022
68. Watts EL, Saint-Maurice PF, Doherty A, Fensom GK, Freeman JR, Gorzelitz JS, et al. 2023. Association of accelerometer-measured physical activity level with risks of hospitalization for 25 common health conditions in UK adults. *JAMA Netw. Open*. 6(2):e2256186. https://doi.org/10.1001/jamanetworkopen.2022.56186
69. Colpitts BH, Mayo A, Bélanger M, Bouchard DR, Boudreau J, Rioux BV, Sénéchal M. 2021. Exploring the association of a total physical activity energy expenditure and diabetes mellitus in adults: a cross-sectional analysis of the Canadian Health Measures Survey. *Metab. Syndr. Relat. Disord.* 19(10):556-561. https://doi.org/10.1089/met.2021.0024
70. Hoyt AL, Rhodes RE, Hausenblas HA, Giacobbi PR. 2009. Integrating five-factor model facet-level traits with the theory of planned behavior and exercise. *Psychol. Sport Exerc*. 10(5):565–572. https://doi.org/10.1016/j.psychsport.2009.02.008
71. Verplanken B, Orbell S. 2003. Reflections on past behavior: a self-report index of habit strength. *J. Appl. Soc. Psychol*. 33(6):1313–1330. https://doi.org/10.1111/j.1559-1816.2003.tb01951.x
72. Gardner B, Abraham C, Lally P, de Bruijn GJ. 2012. Towards parsimony in habit measurement: testing the convergent and predictive validity of an automaticity subscale of the Self-Report Habit Index. *Int. J. Behav. Nutr. Phys. Act*. 9:102. https://doi.org/10.1186/1479-5868-9-102
73. St-Denis B, Beaudry S, Boisgontier MP, Cheval B, Maltagliati S. 2024. Validation de la version francophone de l'Échelle d'Effort Physique. *Mov. Sport Sci. Sci. Mot*. 126(4):33–45. https://doi.org/10.1051/sm/2024019
74. Cheval B, Maltagliati S, Courvoisier DS, Marcora S, Boisgontier MP. 2024. Development and validation of the physical effort scale (PES). *Psychol. Sport Exerc*. 72:102607. https://doi.org/10.1016/j.psychsport.2024.102607
75. R Core Team. 2024. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria (Version 4.4.2). [Computer Software]. https://www.R-project.org/
76. Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using lme4. *J. Stat. Softw*. 2015;67(1):1–48. https://doi.org/10.18637/jss.v067.i01
77. Kuznetsova A, Brockhoff PB, Christensen RHB. lmerTest package: tests in linear mixed effects models. *J. Stat. Softw*. 2017;82(13):1–26. https://doi.org/10.18637/jss.v082.i13
78. O'Brien RM. 2007. A caution regarding rules of thumb for variance inflation factors. *Qual. Quant.* 41(5):673–690. https://doi.org/10.1007/s11135-006-9018-6
79. Bender R, Lange S. 2001. Adjusting for multiple testing—when and how? *J. Clin. Epidemiol*. 54(4):343–349. https://doi.org/10.1016/S0895-4356(00)00314-0

80. Glickman ME, Rao SR, Schultz MR. 2014. False discovery rate control is a recommended alternative to Bonferroni-type adjustments in health studies. *J. Clin. Epidemiol*. 67(8):850–857. https://doi.org/10.1016/j.jclinepi.2014.03.012
81. Benjamini Y, Hochberg Y. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B Stat. Methodol*. 57(1):289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x
82. Akaike H. 1974. A new look at the statistical model identification. *IEEE Trans. Automat. Contr.* 19(6):716–723. https://doi.org/10.1109/TAC.1974.1100705
83. Schwarz G. 1978. Estimating the dimension of a model. *Ann. Stat*. 6:461–464. https://doi.org/10.1214/aos/1176344136
84. Burnham KP, Anderson DR. 2002. *Model selection and multimodel inference: a practical information-theoretic approach*. 2nd ed. New York (NY): Springer. 488 p.
85. Kass RE, Raftery AE. 1995. Bayes factors. *J. Am. Stat. Assoc*. 90(430):773–795. https://doi.org/10.1080/01621459.1995.10476572
86. Wagenmakers EJ, Farrell S. 2004. AIC model selection using Akaike weights. *Psychon. Bull. Rev*. 11(1):192–196. https://doi.org/10.3758/BF03206482
87. Jabouille F, Park K, Boisgontier M. 2026. RR_Validation_KAAT. GitHub. https://github.com/Boisgontier-Lab/RR_Validation_KAAT
88. Allen L, O'Connell A, Kiermer V. 2019. How can we ensure visibility and diversity in research contributions? How the Contributor Role Taxonomy (CRediT) is helping the shift from authorship to contributorship. *Learn. Publ*. 32(1):71–74. https://doi.org/10.1002/leap.1210

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TABLES

Table 2. Study design

Question	Hypotheses	Sampling plan	Analysis Plan	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	Interpretation given different outcomes
What is the construct validity of the KAAT?	To assess the structural aspect of construct validity, which examines whether a measure follows expected internal patterns, we hypothesise that raw reaction times in the KAAT will differ between the approach and avoidance conditions for both physical activity (H1a) and sedentary stimuli (H1b). We further hypothesise that relative bias derived from the KAAT, which accounts for interindividual differences in generic approach-avoidance tendencies by accounting for reaction times to neutral stimuli (11), will significantly differ from zero for both physical activity (H1c) and sedentary stimuli (H1d).	For each experimental stimulus category (physical activity or sedentary behavior), we fitted the following linear mixed-effects model using maximum likelihood estimation with deviation coding (−0.5/+0.5): Reaction Time ~ Movement × Stimulus Type + (1 + Movement | Subject) + (1 | Pictogram). Movement reflected the required response direction (approach vs. avoidance), whereas Stimulus Type contrasted experimental stimuli (physical activity or sedentary pictograms) with neutral pictograms. The selected random-effects structure corresponded to the most theoretically justified specification, including by-subject random slopes for Movement (40), while remaining identifiable and free of singular fits. Consistent with current recommendations to balance model complexity, statistical power, and numerical stability (41,42), we retained the most	To test H1, linear mixed-effects models will be fitted in the R software environment (75) using the lme4 (76) and lmerTest packages (77). The eight sub-hypotheses (H1a-H1h; Table 1) will be tested using the following model specification (R formula syntax), estimated separately for physical activity and sedentary behaviour: Reaction Time ~ Movement * Stimulus Type * Task+Age+Sex+ Height*Weight+ Handedness + Error Rate + Elapsed Time + Task Order + (1 + Task + Movement * Stimulus Type | Subject) + (1 | Pictogram) Categorical predictors will be deviation-coded (−0.5/+0.5), such that main effects represent effects averaged across the levels of the other interacting factors. Continuous covariates will be standardized (z-scored) prior to analysis to facilitate model convergence and interpretation. Random effects will include intercepts for Subject and Pictogram. For participants, random slopes will be specified for Task and for the Movement × Stimulus Type effect, allowing individual differences in overall task effects and approach-avoidance patterns. In the case of non-convergence or singular fit, model simplification will follow a predefined sequence. First, correlations between random slopes and intercepts will be removed. Second, random slopes associated with higher-order interaction terms will be sequentially removed. As a final step, non-essential covariates may be removed, while	Because the interaction effects, such as relative bias, are generally more difficult to detect than simple effects, such as raw bias, we powered the study to detect this interaction. This approach provides a conservative sample-size estimate for H1a and H1b while directly targeting the primary effect interest for H1c and H1d. In linear mixed-effects models, classical power analyses often yield biased or overly optimistic estimates because they do not account for hierarchical variance components and item-level variability (36,37). Therefore, our primary power analysis was conducted using Monte-Carlo simulations with the simr package in R (38), which is specifically designed for power estimation in mixed-effects models. Because no directly comparable effect size estimates were available in the literature for the effects under investigation here, we derived the model	Within each model (physical activity or sedentary): If the simple effect of movement direction is significant, this will be interpreted as evidence of a raw approach-avoidance effect. If the simple effect of movement direction is not significant, this will be interpreted as a lack of evidence for a raw approach-avoidance effect. If the Movement × Task interaction is significant, this will be interpreted as evidence that the magnitude of the raw approach-avoidance bias depends on the type of task used. Specifically, a positive interaction effect will be interpreted as evidence that the KAAT elicits larger approach-avoidance effects than the manikin task, whereas a negative interaction effect will be interpreted as the opposite. If the Movement × Task interaction is not significant, this will be interpreted as a lack of evidence that the raw approach-avoidance bias differs between the KAAT and the manikin task. In this case, both tasks will be considered to potentially capture approach-avoidance tendencies to a comparable extent.

	To assess the sensitivity aspect of construct validity, which examines whether a measure demonstrates enhanced sensitivity to the underlying construct as theoretically expected, we hypothesise that approach-avoidance differences in reaction times, reflecting raw bias, will be larger when assessed with the KAAT than with the manikin task for both physical activity (H1e) and sedentary stimuli (H1f). We formulated analogous hypotheses for relative bias (H1g, H1h).	complex convergent structure supported by the data. Fixed-effect estimates, random-effect variance components (VarCorr), and residual variance (σ^2) from these pilot models were then used to generate simulated datasets reproducing the planned balanced experimental design for each experimental stimulus category (96 neutral trials and 48 experimental trials per participant, with equal numbers of approach and avoidance trials; 20 neutral and 5 experimental stimuli sampled within participants). Statistical power for the Movement × Stimulus Type interaction was estimated using 1000 Monte-Carlo simulations (α = .05; t-test with Satterthwaite-approximated degree of freedom on the fixed effect), yielding a Monte-Carlo standard error below ~1% for the power values around .90 (43). We computed simulation-based power curves using the powerCurve function. These curves indicated that achieving 90% power required 32 participants for physical activity stimuli and 45 participants for sedentary stimuli (Figure 1). We therefore set 45 participants as the minimum required sample size.	all primary fixed effects and their interactions will be retained. This predefined simplification strategy aims to balance model complexity with numerical stability and statistical power (42). For the hypotheses testing internal structure focusing on the KAAT, H1a and H1b (KAAT raw bias) will be tested using the simple effect of Movement (approach vs. avoidance), evaluated at the level of Task = KAAT and with Stimulus Type evaluated at the level of experimental stimuli (physical activity for H1a and sedentary behaviour for H1b). This parameter reflects the raw approach-avoidance bias for physical activity (H1a) and sedentary stimuli (H1b). H1c and H1d (KAAT relative bias) will be tested using the Movement × Stimulus Type interaction, evaluated at Task = KAAT (see section Raw and Relative Automatic Approach-Avoidance Bias). For the sensitivity hypotheses comparing tasks, H1e and H1f (difference in raw bias between tasks) will be tested using the Movement × Task interaction, evaluated at the level of experimental stimuli. H1g and H1h (difference in relative bias between tasks) will be tested using the Movement × Stimulus Type × Task interaction, which evaluates whether the neutral-corrected bias differs between the KAAT and the manikin task. Models will be estimated using restricted maximum likelihood (REML). Statistical assumptions will be evaluated using residual diagnostics (Q-Q plots and residual-versus-fitted plots). If substantial deviations from normality or homoscedasticity are observed, sensitivity analyses will be conducted using log-transformed reaction times to assess the robustness of the findings. Statistical significance will be evaluated at α = .05 using Satterthwaite-approximated degrees of freedom, and effect sizes with 95% confidence intervals will be reported.	parameters from an open dataset (39) obtained from a prior study (4) that used the same manikin task structure and stimuli as in our study. Powering the two-way Movement × Task interaction testing H1e and H1f and the three-way Movement × Stimulus Type × Task interaction testing H1g and H1h, where Task contrasts the KAAT and the manikin task, was not feasible because the KAAT is novel and reliable estimates for between-task differences are unavailable.	If the Movement × Stimulus Type interaction is significant, this will be interpreted as evidence of a relative approach-avoidance effect. If the Movement × Stimulus Type interaction is not significant, this will be interpreted as a lack of evidence for a relative approach-avoidance effect. If the Movement × Stimulus Type x Task interaction is significant, this will be interpreted as evidence that the magnitude of the relative approach-avoidance bias depends on the type of task used. Specifically, a positive interaction effect will be interpreted as evidence that the KAAT elicits larger approach-avoidance effects, whereas a negative interaction effect will be interpreted as the opposite. If the Movement × Stimulus Type x Task interaction is not significant, this will be interpreted as a lack of evidence that the relative approach-avoidance bias differs between the KAAT and the manikin task. In this case, both tasks will be considered to potentially capture approach-avoidance tendencies to a comparable extent.

What is the convergent validity of the KAAT?	To assess convergent validity, which examines the extent to which different measures intended to assess the same construct are related, we hypothesize that raw bias will show small positive correlations between tasks for both physical activity (H2a) and sedentary stimuli (H2b), consistent with previous studies showing limited convergence between approach-avoidance tasks (13,35). We formulated analogous hypotheses for relative bias (H2c, H2d).	The sensitivity analysis yielded a smallest detectable correlation of $\|r\| = .29$.	To assess convergent validity (H2), Pearson correlations will be computed between KAAT and manikin indices of raw approach-avoidance bias (H2a and H2b) and relative approach-avoidance bias (H2c and H2d), separately for physical activity and for sedentary stimuli. Correlation coefficients will be reported alongside 95% confidence intervals and corresponding p-values. The significance level will be set at $\alpha = .05$. Given prior evidence of limited convergence and substantial variability between approach-avoidance tasks (13,35), results will be interpreted with primary emphasis on effect size estimates and their precision rather than statistical significance alone.	For H2a-d, sensitivity power analysis was conducted using G*Power for a bivariate correlation exact test, assuming a two-tailed $\alpha = .05$, 80% power, and a total sample size of N = 90, corresponding to the number of participants that can feasibly be reached.	Significant positive correlations with a 95% confidence interval entirely above r = .29 will be interpreted as support for convergent validity. Significant positive correlations as well as non-significant correlations with a 95% confidence interval including r = .29, will be interpreted as statistically inconclusive, acknowledging limited sensitivity and the need for bigger replication studies. Non-significant correlations with a 95% confidence interval entirely below r = .29 will be interpreted as evidence that any association is smaller than the smallest detectable effect size, providing limited support for convergent validity. Significant correlations in the opposite direction will be interpreted as indicating inconsistency between tasks, calling into question their comparability
What is the concurrent validity of the KAAT?	To assess concurrent validity, which examines the extent to which a measure is associated with theoretically related constructs assessed at the same time, we hypothesise that the relative bias towards physical activity derived from the KAAT will be associated with (H3a) device-based total physical	For H3a-f, the regression models testing included eight predictors: age, sex, height, weight, handedness, total number of errors, time on the task, and the KAAT relative bias term, with the latter being the sole predictor of interest. Sensitivity was estimated as the additional variance explained (ΔR^2) by the KAAT relative bias term when added to the model. For H3g-l, the two task-specific relative biases were reparametrized into their	Concurrent validity will be assessed by examining whether KAAT-derived relative approach-avoidance bias is associated with theoretically related behavioral and motivational indicators measured at the same time point (H3a-H3f), and whether these associations are stronger than those observed for the manikin task (H3g-H3l). For H3a-H3f, separate multiple regressions will be estimated for each outcome variable: device-based total physical activity, device-based moderate-to-vigorous physical activity, explicit affective attitude towards physical activity, intention to be physically active, automaticity towards physical activity, and valuation of physical effort. Each model will include age, sex, height, weight, handedness,	For H3a-l, the sensitivity power analyses were designed to align directly with the planned hypothesis tests. Under the same assumptions as for H2 ($\alpha = .05$, power = .80, N = 90), a priori sensitivity analyses were conducted using G*Power (F tests; linear multiple regression: fixed model, R^2 increase) for H3a-l. The incremental effect was expressed as Cohen's f^2, which can be used to compute $\Delta R^2 = f^2(1 - R^2_{full})$.	If the standardized coefficient β_{KAAT} or the planned contrast ($\beta_{KAAT} - \beta_{Manikin}$) are significant with a 95% confidence interval entirely above zero, this will be interpreted as evidence that the KAAT relative bias is positively associated with the corresponding outcome and, in the case of the contrast, that it shows stronger associations with this outcome than the manikin relative bias. If β_{KAAT} or the planned contrast is statistically significant with a confidence interval entirely below zero, this will be interpreted as

	activity, (H3b) device-based moderate-to-vigorous physical activity, (H3c) explicit affective attitudes towards physical activity, (H3d) intentions to be physically active, (H3e) valuation of physical effort, and (H3f) automaticity towards physical activity. To compare the concurrent validity of the KAAT and the manikin task, we hypothesise that these associations will be stronger for the KAAT than for the manikin task (H3g-l).	mean (capturing shared variance across tasks) and their difference (KAAT − Manikin). This difference term is algebraically equivalent to the planned coefficient contrast ($\beta_{KAAT} - \beta_{Manikin}$) and therefore directly represents the task-difference effect of interest. Each regression model included nine predictors: age, sex, height, weight, handedness, total number of errors, time on the task, the mean bias term, and the bias difference term, with the latter specified as the sole predictor of interest. Sensitivity was estimated as the additional variance explained (ΔR^2) by the bias difference term when added to the model. The smallest detectable incremental effect was $f^2 = 0.089$, which corresponds to a detectable increase in explained variance of approximately $\Delta R^2 = 0.05–0.08$ when assuming plausible full-model R^2 values between 0.10 and 0.40. Because ΔR^2 equals the squared semi-partial correlation of the predictor of interest, this corresponds to a detectable semi-partial association of approximately $r = 0.23–0.28$. In the absence of strong multicollinearity among predictors, standardized regression coefficients (β) are expected	error rate, and elapsed time as covariates, with KAAT relative bias as the predictor of interest. The association between KAAT relative bias and each outcome will be evaluated using the standardized regression coefficient (β), its 95% confidence interval, and corresponding p-value. For H3g-H3l, separate multiple regressions will include the same covariates, along with both KAAT relative bias and manikin relative bias. Including both biases in the same model allows estimation of their unique (partial) associations with each outcome while accounting for shared variance. Standardized regression coefficients and 95% confidence intervals will be reported for each task-specific bias. To formally test whether the KAAT relative bias is more strongly associated with each outcome than manikin bias, planned contrasts comparing regression coefficients ($\beta_{KAAT} - \beta_{manikin}$) will be conducted within each model. A significant positive contrast will indicate that the KAAT-derived bias explains more variance in the outcome than the manikin-derived bias. All continuous covariates and predictors will be standardized prior to analysis. Multicollinearity will be assessed using variance inflation factors (VIF), with values above 5 considered indicative of potentially problematic collinearity. If detected, this will be transparently reported and considered in the interpretation of regression estimates rather than addressed through automatic variable exclusion (78). To control for multiple comparisons within H3a–H3f and H3g–H3l (79), the six hypothesis tests in each set will be treated as a single family. P-values will be adjusted using the Benjamini-Hochberg False Discovery Rate procedure (q = .05) (80,81). Device-based total physical activity will be specified a priori as the primary outcome within each family and will be evaluated without correction.		evidence of an association in the opposite direction. β_{KAAT} or the planned contrasts with a 95% confidence interval including zero will be interpreted as statistically inconclusive. When the 95% confidence interval also spans the minimum detectable effect (approximately β = 0.23-0.28, depending on the full-model R^2), this will indicate limited sensitivity and the need for larger replication studies to conclude on the real effect. Absolute non-significant coefficient (or contrast) with a 95% confidence interval entirely below the minimal detectable effect will be interpreted as evidence that any association or difference between tasks are smaller than the smallest detectable effect size, providing limited support. For secondary outcomes, significant level will be adjusted using the Benjamini-Hochberg false discovery rate procedure (q = .05); the primary outcome (i.e., device-based total physical activity) will be evaluated without correction.

		to be of comparable magnitude.			
What is the predictive validity of the KAAT?	To assess predictive validity, which refers to the extent to which a measure predicts relevant behavioural outcomes, we hypothesise that relative bias towards physical activity stimuli measured with the KAAT will explain variance in device-based total physical activity (H4a), and that kinematic variables (i.e., time to peak velocity, x-flips, and opposite displacement) will explain additional variance beyond that accounted for by relative bias of the KAAT (H4b). We further hypothesise that the relative bias towards physical activity stimuli measured with the KAAT will explain more variance than when assessed with the manikin task (H4c).	For H4a, both the predictor of interest and the total number of predictors were identical to those in H3a-f, yielding a minimal detectable incremental effect size of $f^2 = 0.089$. For H4b, three predictors (x-flips, opposite displacement, and time to peak velocity) were entered at the final step of the model, resulting in 11 total predictors. The minimum detectable incremental effect size was $f^2 = 0.127$. Expressed in terms of explained variance, these values correspond to detectable increases in ΔR^2 of approximately 6–8% for H4a and 8–11% for H4b, when assuming plausible full-model R^2 values between 0.10 and 0.40. Because testing H4c requires comparing non-nested models using information criteria, standard a priori power or sensitivity analyses based on null-hypothesis significance testing are not applicable. Model comparison will therefore rely on relative fit indices (i.e., Akaike Information Criterion [AIC] and Bayesian Information Criterion [BIC]) rather than formal significance testing.	For H4a and H4b, a multiple linear regression model will be estimated in three sequential steps. In Step 1, age, sex, height, weight, handedness, error rate, and elapsed time will the entered as covariates. In Step 2, KAAT relative bias for physical activity will be added. In Step 3: kinematic variables derived from the KAAT (time to peak velocity, number of x-flips, opposite displacement) will be added. H4a will be evaluated by examining the change in explained variance (ΔR^2) when KAAT relative bias is added at Step 2. A statistically significant increase in ΔR^2 at this step will indicate that KAAT relative bias explains variance in device-based total physical activity beyond covariates. H4b will be evaluated by examining ΔR^2 when the kinematic variables are added at Step 3. A statistically significant increase in ΔR^2 at this step will indicate incremental predictive validity of the kinematic variables beyond relative bias. Changes in explained variance will be tested using F-tests for nested models. Standardized regression coefficients (β), 95% confidence intervals, and ΔR^2 values will be reported. Multicollinearity will be assessed as described for H3. To test H4c, a separate regression including the same covariates as in Step 2 will be estimated, replacing the KAAT-derived relative bias with the manikin-derived relative bias. The predictive performance of the KAAT-based model (Step 2) and the manikin-based model will then be compared. Because these models are not nested, comparisons will rely on adjusted R^2 and information criteria (AIC and BIC) (82,83) rather than F-tests. Lower AIC and BIC values indicate better model fit. Differences in AIC (ΔAIC) will be interpreted using established guidelines (84),	For H4a-b, similar G*Power sensitivity analysis parameters used for H3a-l were applied to determine the smallest detectable incremental effect sizes (ΔR^2) for the sequential linear regression models. Standard a priori power or sensitivity analyses based on null-hypothesis significance testing are not applicable for H4-c.	If the addition of the predictors results in a statistically significant increase in explained variance (ΔR^2), this will be interpreted as evidence of predictive validity. Non-significant ΔR^2 changes will be interpreted as statistically inconclusive. When the 95% confidence interval around ΔR^2 includes the minimum detectable effect size (approximately $\Delta R^2 = 0.06–0.08$ for H4a and $\Delta R^2 = 0.08–0.11$ for H4b, depending on the full-model R^2), this will indicate limited sensitivity and the need for larger replication studies. Non-significant ΔR^2 changes with 95% confidence interval entirely below the minimum detectable effect size will be interpreted as evidence that any predictive contribution is smaller than the smallest detectable effect. For H4c, if model K shows better fit than models M (i.e., lower AIC and BIC values), this will be interpreted as evidence of superior predictive validity of the relative bias of the KAAT compared to that of the manikin. Comparable model fit (i.e., ΔAIC or ΔBIC < 2) will be interpreted as suggesting similar predictive

			with values ≤ 2 indicating similar support for both models, values between 4 and 7 indicating less support for the model with the higher AIC, and values > 10 indicating no support for the model with the higher AIC. Differences in BIC (ΔBIC) will be interpreted using established guidelines (85), with values < 2 indicating negligible evidence, values between 2 and 6 indicating positive evidence, values between 6 and 10 indicating strong evidence, and values > 10 indicating very strong evidence against the model with the higher BIC. In addition, Akaike weights will be computed to quantify the relative support for each model, expressed as a probability-like measure of model plausibility (84,86).		contribution of relative bias of both tasks. Better model fit for model M will be interpreted as evidence against the hypothesized predictive validity advantage of the KAAT.